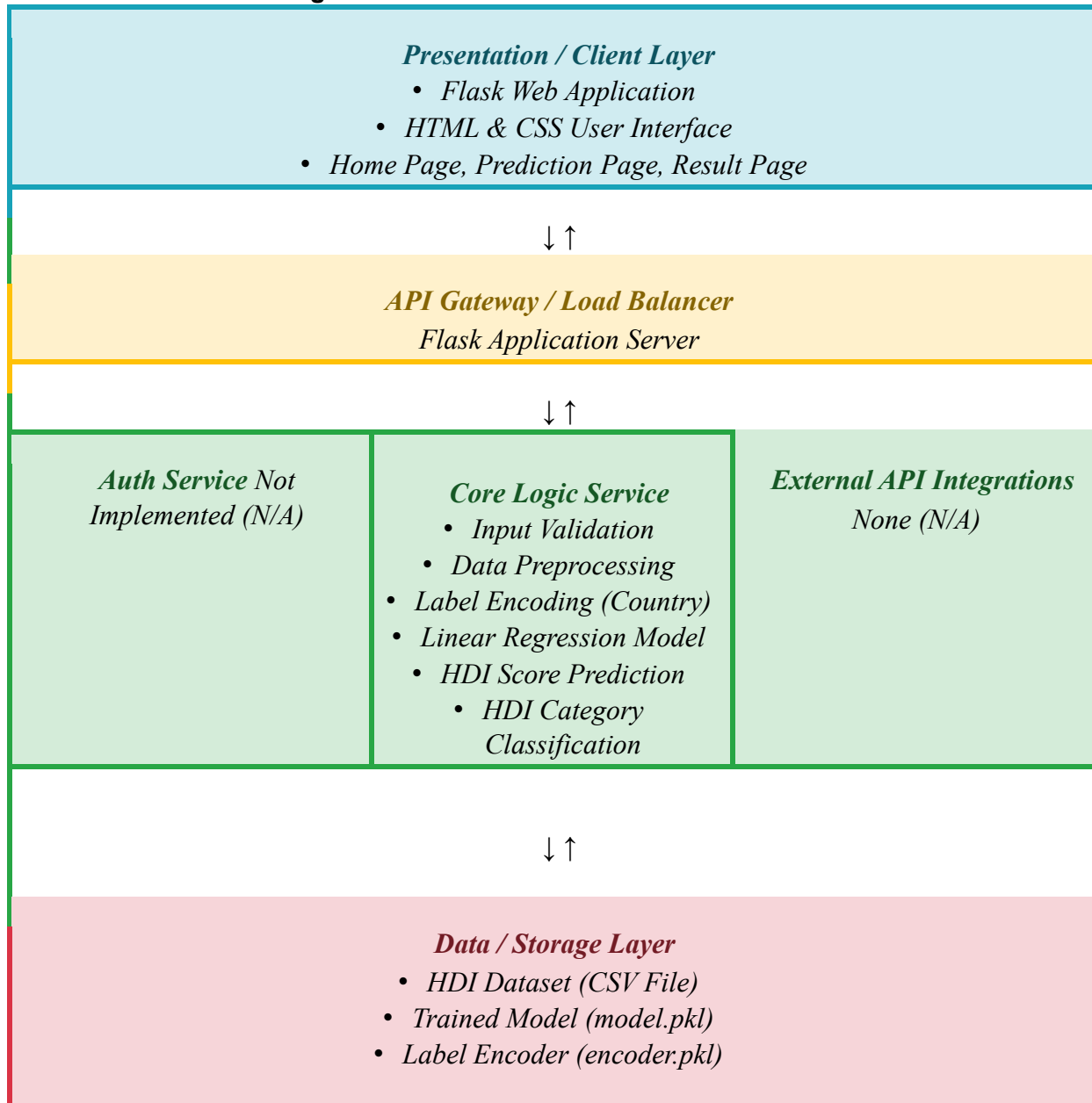


Solution Architecture

Date	25 June 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	Provides a user-friendly interface for entering development indicators and displaying the predicted HDI score and development category.	<i>Flask, HTML, CSS</i>
<i>Application Layer</i>	Receives user input, validates the data, loads the trained Linear Regression model, processes the input, and generates HDI predictions.	<i>Python, Flask, Scikitlearn, Pickle</i>
<i>Core Prediction Service</i>	Performs data preprocessing, encodes the country feature, predicts the HDI score using the Linear Regression model, and classifies the development level.	<i>Python, Scikit-learn</i>
<i>Data Storage</i>	<i>Stores the Human Development Index dataset, trained machine learning model, and label encoder used during prediction.</i>	<i>CSV (HDI Dataset), model.pkl, encoder.pkl</i>